Subject

94693 - Big Data Engineering

8 Credit points

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Subject description

This subject equips you to harness Big Data, transcending traditional analytics with distributed systems, the Hadoop ecosystem, stream processing, and large-scale computation. As a data scientist, your ability to extract value from data aligns with your company's architecture and infrastructure; knowledge of Big Data Engineering enhances this capability. This elective gives you a competitive advantage, preparing you to meet the growing demand for data scientists skilled in Big Data Engineering.

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Faculty
TD School

Subject type
Subject

Study level
Postgraduate

Result type
Grade and marks

Total Workload hours
200

Apply now

Important dates

Fee calculator

Availabilities

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Learning outcomes

Collapse all

Upon completion of this subject, graduates will be able to:

1. Participate in the development of data engineering projects using popular tools (Snowflake, Databricks, PostgreSQL, Spark, Airflow, Kafka etc..) and languages including Python and SQL.
2. Build simple data pipelines or ETLs in order to load, transform and prepare raw data into modern data warehousing solutions.
3. Interact with and query data from modern data warehousing and data lake technologies.
4. Understand basic data engineering concepts including data modelling, big data, distributed data processing and data streaming.

Learning and teaching activities

This subject is conducted over 8 sessions with exercises assigned between classes.
Classes will be conducted in the format of workshop with a mix of lecture components and practical exercises.
Each session runs for 3 hours on Tuesday evenings as decided in the timetable.

The practical components involve two types of activities:

- 'code together' sessions in which the instructor and students build understanding through collaboratively coding solutions to problems or implementing theoretical concepts.
- Practical coding tasks for students to complete themselves or in small groups.

Assignments are a mix of practical coding exercises, solution design and implementation tasks. Through these students get an exposure to historical and current industry trends and challenges, while developing tangible skills to implement these technologies in a work context.

Due to the rapidly advancing nature of this field it is important for students to develop skills in quickly absorbing, dissecting and understanding new technologies and their value to business problems. These assignments in the subject are designed to help students develop these critical new skills.

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Assessment information

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Requisites

Requisite detail

Before you enrol in this subject, you must ensure you have met its requisites. The requisites for this subject can be viewed on its access conditions page.

To learn more about requisites, please visit the How to use the Handbook guide.

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